Università degli Studi di Padova

Department of Physics and Astronomy "Galileo Galilei"
Master Thesis in Physics of Data

Intensive Care Outcome Prediction with Machine Learning

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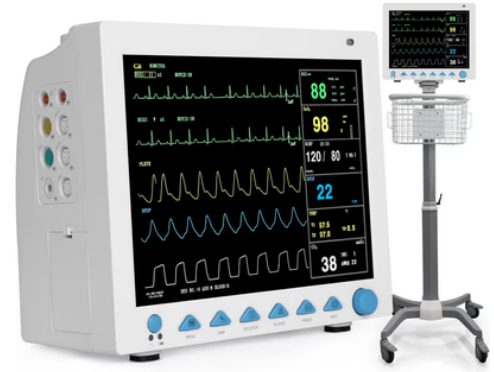
Ventilation Weaning

- is a life saving technique to save patients experiencing acute respiratory failure in Intensive Care Units
- carries risk associated to both premature and delayed extubation
- medical specialists use expertise and structured clinical protocols to find the optimal tradeoff between the risks
- Spontaneous Breathing Trials (SBTs) test the patient's ability to breathe on their own by reducing ventilation provided
- high dimensionality, high quantity, and high complexity of ICU data can lead to making mistakes or missing crucial information
- this results in a current extubation rate of over 15%

Machine Learning

- can solve the current human issues in ventilation weaning
- effortlessly analyzes high dimensional data
- can capture complex, non-linear relationships in a dataset
- binary classification problem to predict the outcome (Success/Failure) of Spontaneous Breathing Trials
- inputs are the pre-SBT windows lasting 15 minutes
- goal:
  - support clinical decision making
  - improve extubation success rate

Dataset



- general ICU population
- timeseries dataset from bedside monitoring machines (standard in ICUs for easy future implementation)
- short pre-SBT windows (15 minutes)
- only the most important physiological signals (11/218 available)

Name	Unit/Format	Meaning	Abundance
P0.1	cmH_2O	airway obstruction pressure at 100 ms	48.08%
RSBI	(breaths/min)/L	rapid shallow breathing index	31.72%
Art-S	mmHg	arterial systolic pressure	90.57%
SpO_2	%	peripheral oxygen saturation	95.65%
SpO_2 PI	%	perfusion index	95.75%
HR	beats/min	heart rate	96.27%
Cdyn	ml/cmH_2O	dynamic compiance	52.52%
PIP	cmH_2O	peak inspiratory pressure	54.91%
fspn	breaths/min	spontaneous respiratory frequency	54.91%
VTe	ml	expiratory tidal volume	54.91%
WOB	J/L	work of breathing	52.61%

- substantially larger dataset, with 1234 SBTs extracted over 1.5 years

Data Preprocessing

- we addressed:
  - missing patient identifiers
  - fragmented ICU stays
  - overlap in records for same stay
- located and filled gaps by interpolating based on the estimated sampling frequency
- discovered redundancies (linear dependence) in medical signals with PCA, discarded VTe and reduced to 10 medical signals
- remove noise and extreme values with moving average filtering and percentile-based clipping (1%-99%)
- discarded stays with insufficient duration for SBT (30 minutes)

SBT Reconstruction

- no annotations available for start, end, or outcome
- start was located as Ventilator Work Mode='PS-CPAP'
- end was determined as min(30 minutes, time of failure)
- literature failure condition: RSBI stabilizes over 105 breaths/min/l
- practical failure condition: 8/10 previous measurements was chosen to match SBT success rate (82.5%)

ML Model

- simple Feed Forward Neural Network to focus on feature engineering
- train (80%) & validation (10%) sets were artificially balanced to help learning despite class imbalance (83% Succeses vs 17% Failures)
- test (10%) set kept unbalanced to quantify real world behaviour
- k-fold cross validation to estimate best hyperparameters

- early stopping based on validation set, weight decay & dropout

Feature Engineering

- standardization to prevent variance sensitivity and help learning
- handcrafted features:
  - mean, standard deviation, skewness
  - 10 signals x 3 features/signal = 30 input features
- FFNN hyperparameters were set via GridSearch based on handcrafted features and used for all strategies

hyperparameter	values	optimal value
hidden layer(s) shape	8, 16, 32, 64, (8,4), (16,8), (32, 16), (32,8)	64
batch size	8, 16	16
weight decay	0.001, 0.0001, 0.00001	0.001
dropout probability	0, 0.1	0.1
activation class	LeakyReLU, ELU	LeakyReLU
learning rate	0.001	0.001
optimizer	Adam	Adam
maximum number of epochs	100	100
early stopping patience	10	10
early stopping threshold	0.00001	0.00001

Feature Engineering

- Principal Component Analysis (PCA) based features
  - dimensionality reduction
  - kept 30 directions corresponding to the most variance
  - still linear transformation
- AutoEncoder (AE) based features
  - unsupervised ML algorithms themselves
  - consist of an encoder and a decoder (symmetrical)
  - once trained, the encoder generates meaningful features
  - Standard AE produces feature representation
  - Variational AutoEncoder (VAE) have a regularization term in loss, predict distribution of features and include a sampling layer
  - convolutional, with similar simple architectures

Results

validation	handcrafted			PCA			AE			VAE		
windows	AUC	prec	rec	AUC	prec	rec	AUC	prec	rec	AUC	prec	rec
right before	0.49	0.54	0.53	0.48	0.84	0.55	0.52	0.82	0.54	0.53	0.84	0.54
all available	X	X	X	0.53	0.84	0.56	0.59	0.83	0.57	0.49	0.83	0.60
test	handcrafted			PCA			AE			VAE		
windows	AUC	prec	rec	AUC	prec	rec	AUC	prec	rec	AUC	prec	rec
right before	0.65	0.96	0.23	0.70	0.92	0.67	0.80	0.91	0.81	0.69	0.92	0.85
all available	X	X	X	0.72	0.94	0.61	0.64	0.89	0.81	0.69	0.82	0.75

- the possible implementations of feature extraction are using:
  - the window right before the start of the SBT
  - all available non-overlapping windows before the SBT start

to test the hypothesis of stationarity of the pre-trial period

Results

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right before	0.49	0.54	0.53	0.48	0.84	0.55	0.52	0.82	0.54	0.53	0.84	0.54
all available	X	X	X	0.53	0.84	0.56	0.59	0.83	0.57	0.49	0.83	0.60
test	handcrafted			PCA			AE			VAE		
windows	AUC	prec	rec	AUC	prec	rec	AUC	prec	rec	AUC	prec	rec
right before	0.65	0.96	0.23	0.70	0.92	0.67	0.80	0.91	0.81	0.69	0.92	0.85
all available	X	X	X	0.72	0.94	0.61	0.64	0.89	0.81	0.69	0.82	0.75

- handcrafted features
  - yield the weakest results (near random) due to limited discriminating power
  - show extreme precision-recall imbalance, consistently with a very conservative decision boundary

Results

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windows	AUC	prec	rec	AUC	prec	rec	AUC	prec	rec	AUC	prec	rec
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all available	X	X	X	0.72	0.94	0.61	0.64	0.89	0.81	0.69	0.82	0.75

- PCA features:
  - produce better and more stable results
  - suggest that linear compression can extract useful features

Results

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right before	0.49	0.54	0.53	0.48	0.84	0.55	0.52	0.82	0.54	0.53	0.84	0.54
all available	X	X	X	0.53	0.84	0.56	0.59	0.83	0.57	0.49	0.83	0.60
test	handcrafted			PCA			AE			VAE		
windows	AUC	prec	rec	AUC	prec	rec	AUC	prec	rec	AUC	prec	rec
right before	0.65	0.96	0.23	0.70	0.92	0.67	0.80	0.91	0.81	0.69	0.92	0.85
all available	X	X	X	0.72	0.94	0.61	0.64	0.89	0.81	0.69	0.82	0.75

- Autoencoders show the most consistently good performance
  - standard AE with just-before windows are best performing, as they can capture complex behaviours with non-linear activations functions
  - VAE display worse performance than standard AE, probably due to a dominance of the Kullback-Leibler divergence regularization term in their loss (we set β=1)

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all available	X	X	X	0.53	0.84	0.56	0.59	0.83	0.57	0.49	0.83	0.60
test	handcrafted			PCA			AE			VAE		
windows	AUC	prec	rec	AUC	prec	rec	AUC	prec	rec	AUC	prec	rec
right before	0.65	0.96	0.23	0.70	0.92	0.67	0.80	0.91	0.81	0.69	0.92	0.85
all available	X	X	X	0.72	0.94	0.61	0.64	0.89	0.81	0.69	0.82	0.75

- single window consistently outperform all windows
  - most recent data is the most representative of current state
  - all windows are non-stationary due to noise from patient care
  - SBT reconstruction is only approximation
- better performance on test set than on validation set : what learned during training (on artificially balanced dataset) works better on the unbalanced test set with binary threshold tuning for each case

Conclusions

validation	handcrafted			PCA			AE			VAE		
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all available	X	X	X	0.53	0.84	0.56	0.59	0.83	0.57	0.49	0.83	0.60
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all available	X	X	X	0.72	0.94	0.61	0.64	0.89	0.81	0.69	0.82	0.75

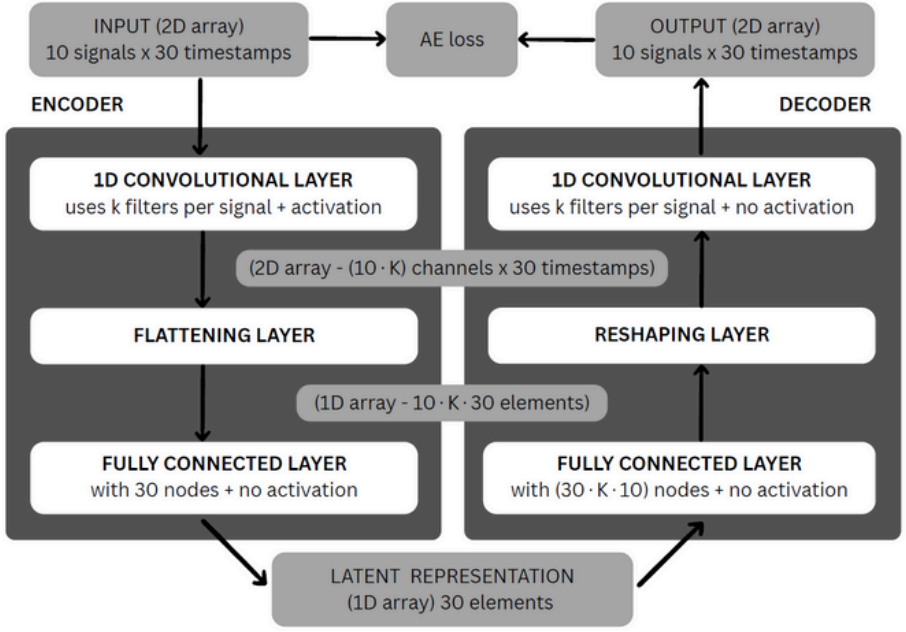
- showed that data representation can strongly influences predictive performance
- Standard AE led to best performing machine learning model
- proved that:
  - short pre-SBT windows are effective for predicting SBT outcome
  - models based on timeseries data extracted from standard ICU data can be effective for predicting SBT prediction

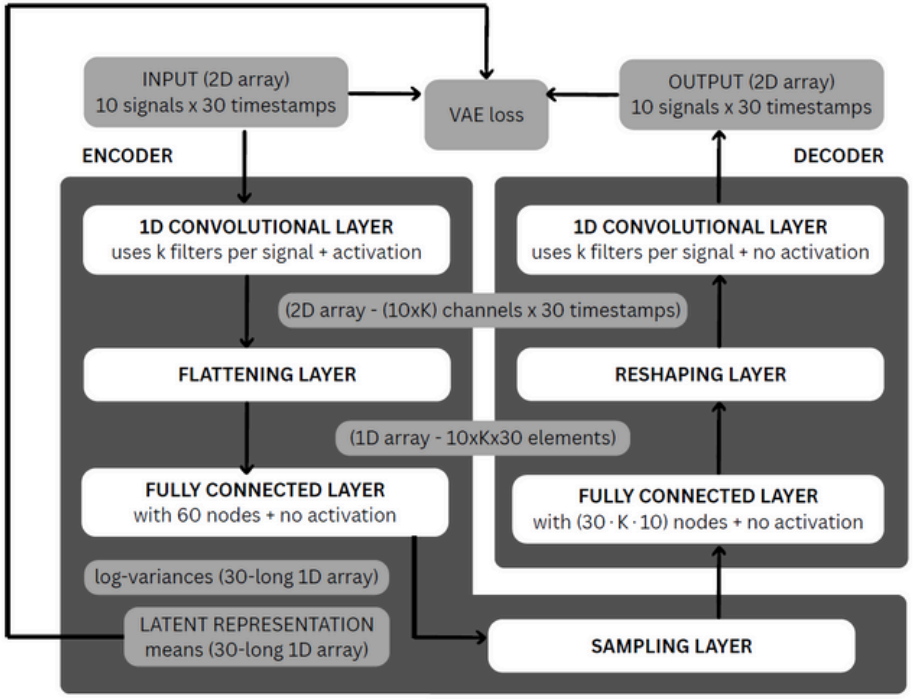
Future Prospects

- retrain all models without balancing training and validation sets
- analyze difference in single-vs-all windows distributions with Kullback-Leibler Divergence
- enrich handcrafted features with temporal descriptors (e.g. gradient)
- further improving VAE with varying β or de-noising
- SHAP value analysis to explain contribution of each feature
- more annotations and metadata
- using other ICU datasets

Thank you
for your attention!

AE vs VAE Architectures





AE vs VAE Hyperparameters

hyperparameter	values	best AE value (all windows)	best AE value (just before)
batch size	64, 128	64	64
weight decay	0.00001, 0.0001	0.00001	0.0001
dropout probability	0, 0.1	0	0
activation class	LeakyReLU, ELU	ELU	LeakyReLU
learning rate	0.00003, 0.0001	0.001	0.001
optimizer	Adam	Adam	Adam
loss function	MSE, MAE	MSE	MSE
filters per signal k	2, 3, 4	4	4
filter size	3, 5, 7	3	3
maximum number of epochs	100	100	100
early stopping patience	10	10	10
early stopping threshold	0.0001	0.0001	0.0001

hyperparameter	values	best VAE value (all windows)	best VAE value (just before)
batch size	64, 128	128	64
weight decay	0.00001, 0.0001	0.00001	0.00001
dropout probability	0, 0.1	0	0
activation class	LeakyReLU, ELU	LeakyReLU	ELU
learning rate	0.0003, 0.001	0.001	0.001
optimizer	Adam	Adam	Adam
loss function	MSE, MAE	MSE	MSE
filters per signal k	2, 3, 4	4	4
filter size	3, 5, 7	7	5
maximum number of epochs	100	100	100
early stopping patience	10	10	10
early stopping threshold	0.0001	0.0001	0.0001
KL-divergence weight β	1	1	1